

The constructionalization of movement repetition: Explaining noun-verb pairs in sign languages

Noun-verb pairs are widely attested in many unrelated sign languages where two signs share core semantic and phonological features but differ in movement, including movement repetition, the focus of this paper due to its identifiability. Take TAKE-PICTURE/CAMERA in American Sign Language (ASL); short repeated movements characterize the noun while the non-iterative verb contains a singular movement. Current theoretical accounts include root-derivation via an underlying form (Supalla & Newport 1978) and stem-derivation via nominalizing reduplication, a derivational process where the movement of a verb is repeated to derive a related nominal meaning (Abner 2017). However, these accounts cannot explain the unpredictability of the resulting semantics and partial productivity of noun-verb pairs, and, importantly, why movement repetition presents in multiple unrelated sign languages as this particular syntactic categorical distinction. Unrelated sign languages frequently share similar linguistic properties because they recruit iconicity, or the relationship between a sign's form and its meaning, as a substrate for sign creation. Thus, iconicity seems a natural hypothesis for the ubiquity of movement repetition; however, Pyers & Emmorey (2022) show that hearing non-signers inaccurately perceive movement repetition in ASL nominal signs as denoting verbal meanings. Instead of transparently deriving from iconicity, movement repetition appears to have grammaticalized and requires linguistic explanation.

Using Construction Morphology (Booij 2010), Hüning & Booij (2014) demonstrate the *constructionalization*, or the emergence of a constructional schema, of a derivational construction where it becomes more morphemic in form and meaning. I propose a constructionalization of movement repetition from iconic origins to a derivational marker distinguishing between noun and verb signs. The resultant schema is in (1); a reciprocal relationship ($\leftrightarrow$) between the form and meaning accounts for the differences in noun-verb pairs, where movement repetition is appended onto a verb *a* to create a novel nominal sign. Subschemas further capture the varying, unpredictable semantics of the nominal sign via more detailed schemas (e.g. predictable nominal meanings) to particular semantic families, citing the semi-productivity of movement repetition.

I propose the constructionalization of movement repetition originates from pluractionality, where a verb's movement is iconically repeated to indicate multiple events, in (2) and its schema in (3). This phrasal construction, typically expressed with three movement instantiations and head movements, is reduced to activity noun signs with two movements, in (4). The agentive noun subschema in (5) likewise recruits pluractionality where the pluractional phrase is embedded in relative clauses to denote individuals: DOCTOR is construed as 'they who take pulses repeatedly.' This embedding can also account for object nouns, which comprise much of ASL's lexicon; CAMERA is an instrument used repeatedly in photography, in (6). Activity and agentive nouns strongly evidence that iconic pluractional modulation leads to the increased use of movement repetition to denote nominal meanings. With frequent use, the iconic relation between form and meaning weakens so that hearing non-signers cannot recruit iconicity to accurately perceive movement repetition. This highlights how constructions with movement repetition can emerge from iconic construals but present non-iconically, which has crucial implications for how sign lexicons emerge and are organized.

- (1) $[[a]_{Vj} + \text{movement repetition}]_k \leftrightarrow [[a]_{Nj}]_k$ 'N with semantic properties of V'
- (2) IX TEND-TO [WRITE WRITE WRITE]_{pluractional} 'They write often.'
- (3) $[[a]_{Vj} + \text{movement repetition} + \text{head mov.}]_k \leftrightarrow [[a]_{V\text{pluractionality}j}]_k$ 'repeated V'
- (4) $[[a]_{Vj} + \text{movement repetition}]_k \leftrightarrow [[a]_{V\text{activity}j}]_k$ 'an activity resulting from repeated V'
- (5) $[[a]_{Vj} + \text{movement repetition}]_k \leftrightarrow [[a]_{V\text{agentive}j}]_k$ 'an agent associated with repeated V'
- (6) $[[a]_{Vj} + \text{movement repetition}]_k \leftrightarrow [[a]_{V\text{object}j}]_k$ 'an instrument associated with repeated V'

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